Information Retrieval Introduction

Lecture 1: Vector Space Model & Boolean Retrieval

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Information Retrieval

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Course Overview

1. **Introduction to IR: Vector Space Model & Boolean Retrieval** ← Today
2. Indexing: Sparse and Dense
3. Query Understanding & Expansion
4. Evaluation of IR Systems
5. Ranking: BM25, Language Models, and Beyond
6. Embeddings & Neural Ranking
7. Dense Retrieval
8. Learning to Rank
9. Recommender Systems
10. Retrieval Augmented Generation (RAG)
11. LLMs as a Judge

Today's Goals:

- Understand what IR is and why it matters
- Learn the fundamental IR process

Outline

1. What is Information Retrieval?
2. Challenges in IR
3. The Information Retrieval Process
4. Boolean Retrieval
5. The Vector Space Model
6. Questions

What is Information Retrieval?

What is Information Retrieval?

Definition

Information Retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need from within large collections (usually stored on computers).

The Essence of IR: Given an Information Need

- **Find relevant results** that satisfy the information need
- **By searching over a large collection** of items (billions of documents)
- **In reasonable time:**
  - Web search: < 1 second
  - AI/conversational systems: allow more time for generation
- **Present results** in a manner that is easily consumable by humans

IR vs. Database Systems

Aspect	Database Systems	IR Systems
Data	Structured (tables, schemas)	Unstructured (text, documents)
Queries	Precise (SQL)	Vague (natural language)
Matching	Exact	Approximate/Ranked
Result	All matching records	Top-k relevant documents
Failure	No results or all results	Ranked list (best effort)
Example	`SELECT * WHERE age>30`	"climate change effects"

Key Difference

Databases: *"Give me all records matching X"*
IR: *"Give me the most relevant documents about X"*

IR vs. Natural Language Processing (NLP)

The Relationship

IR and NLP are closely related but have different focal points:

NLP (Language Centric)

- Focus on language **understanding**, representation, and **generation**.
- Deep analysis of syntax, semantics, and pragmatics.
- Models like BERT, Llama, GPT.

IR (Scale & Need Centric)

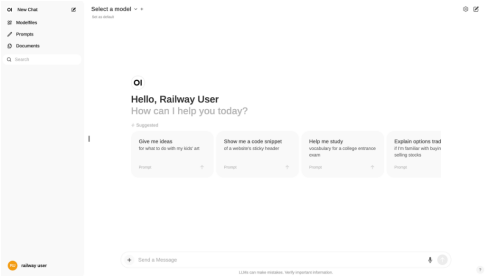
- Focus on **information seeking** at a massive scale.
- Uses NLP tech as a toolkit to model relevance.
- Priority: Speed, ranking, and satisfying user needs.

Typical Tasks

- **NLP:** Named Entity Recognition, Dependency Parsing, Translation, Summarization.
- **IR:** Web Search, Recommendation Systems, Ad-hoc Retrieval, Question Answering.

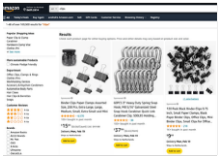
Examples of Search Systems

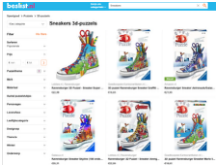
Conversational AI Search



ChatGPT - AI-powered conversational search

E-Commerce Search





Amazon - Product search and recommendations

Common Thread

All search systems share the same core challenge: finding relevant items from massive collections in real-time.

Real-World IR Applications

Web Search

- Google, Bing, DuckDuckGo
- Billions of web pages
- Sub-second response time

Enterprise Search

- Search company documents
- Email search
- Knowledge management

E-commerce

- Amazon product search
- eBay, Etsy
- Recommendation systems

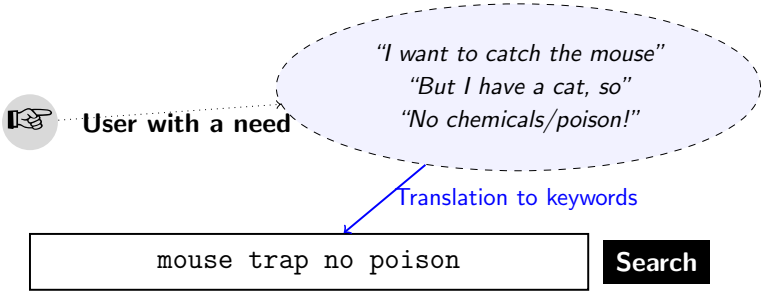
Specialized Domains

- Legal document search (LexisNexis)
- Academic papers (Google Scholar)
- Medical records (PubMed)
- News archives

IR is everywhere you search for information!

Challenges in IR

The Retrieval Challenge: Understanding Intent



The Semantic Gap

- **Syntactic match:** Docs containing "mouse", "trap", "poison".
- **User intent:** Docs describing *humane* traps or *mechanical* traps.
- **Danger:** Returning docs about "how poison traps work" (contains keywords!).

The Scale of Modern IR

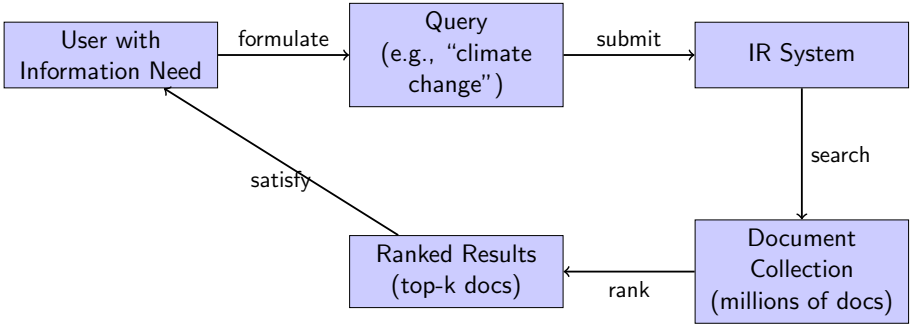
Web Search Statistics (2024)

- **Google:** Processes 8.5+ billion searches per day
- **Index size:** 100+ trillion web pages indexed
- **Freshness:** New pages indexed within hours
- **Speed:** Results in < 0.5 seconds on average

Challenges at Scale

- **Storage:** How to store 100 trillion documents?
- **Speed:** How to search in milliseconds?
- **Relevance:** How to rank billions of results?
- **Freshness:** How to keep index up-to-date?
- **Quality:** How to fight spam and misinformation?

The Information Retrieval Problem



Core Questions:

1. How to represent documents and queries?
2. How to match queries to documents efficiently?
3. How to rank documents by relevance?

The Information Retrieval Process

Forward Reference: Lecture 2

The fundamental IR process, including pipeline architectures and preprocessing, is now covered in detail in **Lecture 2: Indexing**.

Why the change?

- Better logical flow: Preprocessing is tightly coupled with building indexes.
- Focus for today: Understanding the models (Boolean and Vector Space).

Proceeding to Section 3: Boolean Retrieval...

Boolean Retrieval

Boolean Retrieval Model

Definition

Documents are retrieved if they satisfy a Boolean expression of query terms

Boolean Operators:

- **AND:** Both terms must appear (intersection)
- **OR:** Either term must appear (union)
- **NOT:** Term must not appear (negation)

Example Queries

- `climate AND change` → docs containing both "climate" and "change"
- `cat OR dog` → docs containing "cat" or "dog" or both
- `java NOT coffee` → docs with "java" but not "coffee"
- `(climate AND change) OR (global AND warming)`

Boolean Retrieval with Sets

Documents as Sets of Terms

Document Collection:

- D1: {cat, sat, mat}
- D2: {dog, sat, log}
- D3: {cat, dog, run}
- D4: {bird, fly, sky}

Inverted Index:

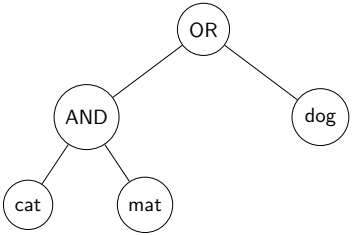
- cat: {D1, D3}
- dog: {D2, D3}
- sat: {D1, D2}
- run: {D3}

Query Processing:

- `cat AND dog`: $\{D1, D3\} \cap \{D2, D3\} = \{D3\}$
- `cat OR dog`: $\{D1, D3\} \cup \{D2, D3\} = \{D1, D2, D3\}$
- `cat NOT dog`: $\{D1, D3\}$ - $\{D2, D3\} = \{D1\}$

Boolean Retrieval Example

Query: `(cat AND mat) OR dog`



Execution:

1. `cat`: {D1, D3}
2. `mat`: {D1}
3. `cat AND mat`: {D1, D3} $\cap$ {D1} = {D1}
4. `dog`: {D2, D3}
5. `(cat AND mat) OR dog`: {D1} $\cup$ {D2, D3} = {D1, D2, D3}

Advantages of Boolean Retrieval

✓ Advantages

- ✓ **Precise control:** Users specify exactly what they want
- ✓ **Transparent:** Clear why documents matched
- ✓ **Efficient:** Fast set operations on posting lists
- ✓ **Predictable:** Same query always returns same results
- ✓ **Expressive:** Can construct complex queries

Use Cases

- Legal document search (very precise requirements)
- Patent search (need exact specifications)
- Database-like queries on text
- Expert users who know what they want

Limitations of Boolean Retrieval

✗ Critical Limitations

✗ **No ranking**

- All matching documents equally relevant
- Can't distinguish "very relevant" from "barely relevant"

✗ **All or nothing**

- `climate AND change AND effects`: No results if one term missing
- Too restrictive or too permissive

✗ **Hard to formulate queries**

- Most users don't know Boolean operators
- Hard to express fuzzy information needs

✗ **Result set size unpredictable**

- Could return 0 results or 10 million
- Users want "top 10", not "all matching"

Boolean Retrieval: The Feast or Famine Problem

Too Restrictive (Famine)

- Query: `machine AND learning AND neural AND networks`
- Result: 0 documents
- Problem: One term missing kills entire result

Solution?

- User tries: `machine AND learning`

Too Permissive (Feast)

- Query: `machine AND learning`
- Result: 1,000,000 documents
- Problem: Which 10 should I read?

Need:

- Ranking by relevance
- Graceful degradation
- Better matching

This motivates the Vector Space Model!

The Vector Space Model

From Boolean to Vector Space

Key Insight

Instead of documents being sets of terms (in/out), represent them as **vectors** with weights for each term

Boolean Model:

- D1: {cat, dog}
- D2: {cat, mat}
- Binary: term present or not

Query: `cat AND dog`

- Result: {D1}
- D2 excluded (no "dog")

Vector Space Model:

- D1: [cat: 0.5, dog: 0.5, mat: 0]
- D2: [cat: 0.7, dog: 0, mat: 0.3]
- Weighted: importance/frequency

Query: [cat: 0.5, dog: 0.5]

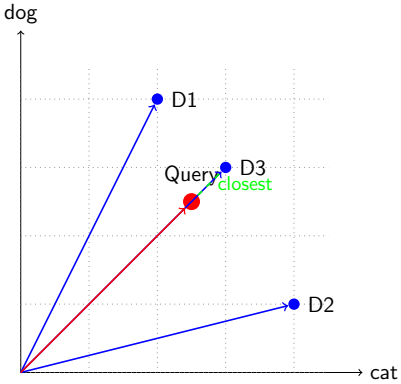
- Similarity(D1, Q) = 0.5
- Similarity(D2, Q) = 0.35
- **Ranked:** D1, then D2

Advantage

D2 still returned (has "cat"), just ranked lower than D1

Vector Space Model: Geometric View

Documents and queries as vectors in term space



Similarity = Angle between vectors
(Smaller angle = more similar)

Term Weighting in Vector Space Model

Question: How to assign weights to terms?

Option 1: Binary Weights

$$w_{t,d} = \begin{cases} 1 & \text{if term } t \text{ appears in doc } d \\ 0 & \text{otherwise} \end{cases}$$

- Simple but ignores frequency

Option 2: Term Frequency (TF)

$$w_{t,d} = \text{count}(t, d)$$

- Better: reflects importance
- Problem: Common words get high weight

Problem with Raw TF

Document mentioning "the" 100 times doesn't make it about "the"

TF-IDF Weighting

Idea: Balance term frequency with term rarity

TF-IDF Formula

$$w_{t,d} = \text{tf}(t, d) \times \text{idf}(t)$$

where:

- $\text{tf}(t, d) = \text{count}(t, d)$ (term frequency in document)
- $\text{idf}(t) = \log \frac{N}{\text{df}(t)}$ (inverse document frequency)
- N = total number of documents
- $\text{df}(t)$ = number of documents containing term t

Intuition:

- **High TF:** Term appears often in this document $\rightarrow$ important for this doc
- **High IDF:** Term is rare across collection $\rightarrow$ discriminative
- **High TF-IDF:** Important AND discriminative

TF-IDF Example

Collection: 10,000 documents

Term	tf(t,d)	df(t)	idf(t)	tf-idf
neural	5	100	$\log(10000/100) = 2.0$	10.0
network	3	500	$\log(10000/500) = 1.3$	3.9
the	50	9000	$\log(10000/9000) = 0.05$	2.5

Observations:

- "neural" appears less than "the" (5 vs. 50) but gets higher weight (10.0 vs. 2.5)
- Rare, discriminative terms weighted higher
- Common words automatically downweighted

Note

This is a simplified version; we'll cover refined TF-IDF and BM25 in Lecture 3

Document Vectors with TF-IDF

Example Collection:

- D1: "cat cat dog"
- D2: "cat mat"
- D3: "dog bird bird"

Vocabulary: {cat, dog, mat, bird}

Doc	cat	dog	mat	bird
D1	$2 \times \log(3/2) = 0.35$	$1 \times \log(3/2) = 0.18$	0	0
D2	$1 \times \log(3/2) = 0.18$	0	$1 \times \log(3/1) = 0.48$	0
D3	0	$1 \times \log(3/2) = 0.18$	0	$2 \times \log(3/1) = 0.95$

Vector Representation:

- D1: [0.35, 0.18, 0, 0]
- D2: [0.18, 0, 0.48, 0]
- D3: [0, 0.18, 0, 0.95]

Cosine Similarity

How to measure similarity between vectors?

Cosine Similarity Formula

$$\text{sim}(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{|\vec{q}| \times |\vec{d}|} = \frac{\sum_t q_t \times d_t}{\sqrt{\sum_t q_t^2} \times \sqrt{\sum_t d_t^2}}$$

Properties:

- Range: [-1, 1] (or [0, 1] for positive weights)
- 1 = identical vectors (same direction)
- 0 = orthogonal (no common terms)
- -1 = opposite vectors (not relevant for IR)

Why cosine instead of dot product?

- Cosine normalizes for document length
- Long documents don't automatically score higher

Cosine Similarity: Example Calculation

Query: "cat dog" $\rightarrow \vec{q} = [0.5, 0.5, 0, 0]$
Documents:

- D1: [0.35, 0.18, 0, 0]
- D2: [0.18, 0, 0.48, 0]
- D3: [0, 0.18, 0, 0.95]

Similarity Calculations:

$$\begin{aligned}
\text{sim}(q, D1) &= \frac{(0.5 \times 0.35) + (0.5 \times 0.18)}{\sqrt{0.5^2 + 0.5^2} \times \sqrt{0.35^2 + 0.18^2}} \\
&= \frac{0.265}{0.707 \times 0.394} = 0.95 \\
\text{sim}(q, D2) &= \frac{0.5 \times 0.18}{0.707 \times 0.511} = 0.25 \\
\text{sim}(q, D3) &= \frac{0.5 \times 0.18}{0.707 \times 0.967} = 0.13
\end{aligned}$$

Ranking: D1 (0.95) ¿ D2 (0.25) ¿ D3 (0.13)

VSM vs. Boolean Retrieval

Aspect	Boolean	Vector Space
Matching	Exact (in/out)	Partial (weighted)
Ranking	No	Yes (by similarity)
Result size	Unpredictable	Top-k
Term weighting	Binary	TF-IDF
Query expression	Boolean operators	Term list
Robustness	Brittle	Graceful degradation
User friendliness	Expert users	General users
Relevance	All equal	Ranked by relevance

Modern Usage

- **Boolean:** Still used in legal, patent search (precision critical)
- **Vector Space:** Foundation for web search, most IR systems
- **Hybrid:** Many systems combine both approaches

Advantages of Vector Space Model

✓Key Advantages

1. **Ranking by relevance**
   - Documents ordered by similarity to query
   - Users get best results first
2. **Partial matching**
   - Documents matching some query terms still retrieved
   - Graceful degradation (no feast/famine)
3. **Term weighting**
   - Important terms weighted higher (TF-IDF)
   - Automatic handling of common words
4. **Simple queries**
   - Users just type terms (no Boolean operators)
   - Natural for web search
5. **Tunable**
   - Can adjust weighting schemes
   - Foundation for more advanced models (BM25, Lecture 3)

Limitations of Vector Space Model

✗ Limitations

1. **Bag of words assumption**
   - Word order ignored: "dog bites man" = "man bites dog"
   - No phrase matching
2. **Term independence**
   - Assumes terms are independent
   - "New York" treated as "New" and "York"
3. **Synonymy**
   - "car" and "automobile" treated as different
   - Misses relevant documents with synonyms
4. **Polysemy**
   - "bank" (financial) vs. "bank" (river) undistinguished
   - Can retrieve irrelevant documents

Looking Ahead